

Introduction to Macroeconomics – August 2022

Question 1 (ch.1)

Which of the following is a macroeconomic rather than a microeconomic phenomenon?

- a. The extent of competition in the dairy market.
- b. The extent to which monetary policy can address medium run unemployment objectives.
- c. The cost structure of a state owned company.
- d. Consumer decisions under risk aversion.

Answer: b)

Question 2 (ch.2)

Given the following information, which statement is true?

Exports (X): 20

Imports (M): 3

Gross domestic product: 100

Net factor income from abroad: 2

Salaries earned by foreigners inside the domestic territory: 3

Indirect taxes: 3

Production subsidies: 2

Savings: 14

Capital depreciation: 10

- a. National income is larger than gross domestic product.
- b. Net domestic product is higher than total added value at market prices.
- c. Gross national product is equal to 103.
- d. National income is equal to 91.

Answer: d

Question 3 (ch.2)

A country produces the following quantities of masks and hydroalcoholic gel at the prices in the table below:

year	Masks prices	Masks quantities	Hydroalcoholic gel prices	Hydroalcoholic gel quantities
2019	1	100	2	150
2020	4	400	5	400

The base year is 2019. Which of the following statements is correct?

- a. Real GDP was multiplied by a factor of ten between 2019 and 2020.
- b. The GDP deflator is equal to 300 in 2020.
- c. The GDP deflator is equal to 30 in 2020.
- d. Real GDP in 2020 is smaller than nominal GDP in 2019.

Answer: b)

Question 4 (ch.2)

Which of the following propositions is correct?

- a. Consumption is the largest component of GDP among high income countries.

- b. Household expenditures when purchasing their homes are included in the consumption aggregate when calculating GDP.
- c. National income is a measure of a country's production at market prices, and therefore a better indicator than GDP of the productive capacity of a country.
- d. GDP takes into account volunteer work.

Answer: a)

Question 5 (ch.3)

Nominal monthly wages were around CHF 1'500 in 1960 and CHF 7'500 in 2020. Between 1960 and 2020, the Consumer Price Index increased from 50 to 100. Which of the following propositions is correct?

- a. Real monthly wages increased 50% between 1960 and 2020.
- b. The purchasing power of CHF 7'500 in 1960 is 2 times higher than in 2020.
- c. The growth rate of nominal wages is smaller than the growth rate of consumer prices between 1960 and 2020.
- d. To keep constant purchasing power between 1960 and 2020, nominal wages would have needed to be around CHF 4'500 rather than CHF 7'500.

Answer: b)

Question 6 (ch.3)

Which of the following propositions is correct?

- a. The GDP deflator and the Consumer Price Index are computed differently: only one of them depends on the choice of a base year.
- b. If consumers consumed the same basket of goods every year, then the Consumer Price Index would be the same as the GDP deflator.
- c. Any variation in the price of satellites produced by the NASA will impact the US Consumer Price Index, even if all the other prices stay constant.
- d. The GDP deflator and the Consumer Price Index are correlated.

Answer: d

Question 7 bis (ch.4)

According to the economic growth model by Solow, which of the following propositions can be stated with certainty?

- a. Economic convergence is an implication of the assumption of constant marginal factor return.
- b. A poor country with a lower endowment of physical capital will eventually have the same economic growth rate than a rich country.
- c. Economic convergence is only possible if there is constant technological progress.
- d. In the presence of unconditional economic convergence, there is limited economic growth.

Answer: b)

Question 8 (ch.4)

If the aggregate production function has diminishing returns, an increase of the labor force will result in an increase in aggregation production and...

- a. An increase in output per worker.
- b. No changes in output per worker.
- c. A decrease in output per worker.
- d. A decrease in output per unit of capital.

Answer: c.

Question 9 (ch.5)

In Switzerland, which of the following criteria is a necessary condition for a worker to be considered as unemployed?

- a. Satisfy the criteria defined by the International Labor Organization.
- b. Not to be working.
- c. Benefit for an unemployment allowance.
- d. Be registered in a placement regional office (Office régional de Placement).

Answer: d.

Question 10 bis (ch.5)

Which of the following propositions regarding the labor market is correct?

- a. Setting a minimum wage necessarily leads to a reduction in employment.
- b. Frictional unemployment will vanish if the wage could fluctuate freely.
- c. It is because of labor unions that firms introduce efficiency wages.
- d. Efficiency wages are higher than equilibrium wages.

Answer: d)

Question 11 (ch.5)

Assuming that the labor market is perfectly competitive, consider the following labor supply and demand functions: $LS = 4w$ and $LD = 50 - w$, where w is the hourly wage. Which of the following propositions is true?

- a. In equilibrium, the marginal productivity of labor is equal to 5.
- b. 25 workers will earn their reservation wage.
- c. Implementing a minimum wage of 12 will not provoke any unemployment.
- d. If the labor supply function increases at $LS = 9w$, the equilibrium wage will decrease by 50% in comparison to the initial equilibrium.

Answer: d

Question 12 (ch.6)

The government of a closed economy decides to reduce taxes on saving, but at the same time, it increases government expenditure. What is the impact on the equilibrium interest rate and quantity of loanable funds?

- a. Both the interest rate and the quantity of loanable funds increases.
- b. The impact on the interest rate is ambiguous and the quantity of loanable funds declines.
- c. Both the interest rate and the quantity of loanable funds remain unchanged.
- d. The impact on both the interest rate and the quantity of loanable funds is ambiguous.

Answer: d.

Question 13 (ch.6)

Due to the COVID pandemic, the government increased its health expenditures, leading to an increase in government borrowing. If we were to focus on the market for loanable funds, which of the following can be stated with certainty?

- a. The increase in government borrowing shifts to the right the demand of loanable funds. The crowding out effect in the market for loanable funds amplifies this shift.
- b. The increase in government borrowing increases the interest rate, but the impact on the quantity of loanable funds is ambiguous.

- c. The increase in health expenditure shifts the supply of loanable funds to the left. The demand of loanable funds remains unchanged.
- d. The increase in health expenditure raises the interest rate and has no impact on the quantity of loanable funds borrowed by the private sector.

Answer: c.

Question 14 (ch.7)

Which of the following propositions is **NOT** one of the three essential functions of money?

- a. Serve as a unit of account.
- b. Serve as a medium of exchange.
- c. Serve to trace monetary transactions.
- d. Serve to transfer purchasing power across time.

Answer: c)

Question 15 (ch.7)

Consider an economy with a reserve requirement $r = 20\%$ and with $M0 = 1600$ and $M1 = 3200$. Which of the following propositions is correct?

- a. People hold approximately 30% of their money in cash.
- b. The money multiplier is equal to 3.
- c. Decreasing the reserve requirement would decrease the money multiplier by 20%.
- d. People hold approximately 50% of their money in cash.

Answer: a

Question 16 bis (ch.13)

The marginal propensity to consume is 0.80 and the marginal propensity to import is 0.30. What would be the increase in aggregate expenditure following an increase in government expenditure of CHF 10'000 CHF?

- a. CHF 20'000.
- b. CHF 11'000.
- c. CHF 50'000.
- d. CHF 10'000.

Answer: a)

Question 17 (ch.8)

According to the quantitative theory of money:

- a. Inflation is the result of unemployment being below the structural level of unemployment
- b. The velocity of money increases in the presence of deflation.
- c. Nominal wages are rigid in the long-run.
- d. Money has no impact on real output.

Answer: d

Question 18 bis (ch.14)

The short run Phillips curve « a » coefficient is equal to 0.2. Expected inflation is 10%, actual inflation is 12% and the natural level of unemployment is 5%. What is the actual level of unemployment?

- a. 4.6%.
- b. 2%.
- c. 5.4%.
- d. 7%.

Answer: a.

Question 19 (ch.9)

Which of the following propositions regarding the balance of payment is correct?

- a. A purchase of foreign government bonds is equivalent to capital imports.
- b. Sales of a domestic firm's shares to a resident in a foreign country is accounted in the capital/financial account. The payment of dividends associated with the foreign ownership of these shares is accounted in the Net Factor Income from Abroad balance.
- c. A surplus of the balance of payment occurs if the net capital outflow balance is negative.
- d. The current account only considers international trade in goods, but not services.

Question 20 bis (ch.9)

Which of the following statements concerning government deficits in an open economy is necessarily correct?

- a. They generate a negative trade balance.
- b. They can be financed by either domestic or foreign loanable funds.
- c. They crowd out private investments only if the trade balance is negative.
- d. They do not affect national savings.

Answer (b)

Question 21 (ch.9)

In an open economy, GDP is equal to 50, investment is equal to 20 and consumption is equal to 10. Net capital outflows in this economy are negative. It can be stated that:

- a. Government expenditure and national savings are both equal to 20.
- b. Government expenditure is greater than 20 and national savings are lower than 20.
- c. Government expenditure and national savings are both lower than 20.
- d. Government expenditure is smaller than 20 and national savings are greater than 20.

Answer (b)

Question 22 (ch.10)

In December 2021 the EUR-CHF exchange rate is 1.046 (i.e. 1EUR = 1.046CHF) and during 2021 it oscillated between 1.0374 and 1.1152, with a peak at 1.1152 in March 2021. From the perspective of Switzerland, which of the following propositions is correct?

- I. The exchange rate is expressed in indirect quotation.
 - II. The exchange rate is expressed in direct quotation.
 - III. Between March and December 2021, the Swiss Franc appreciated with respect to the Euro.
 - IV. Between March and December 2021, the Swiss Franc depreciated with respect to the Euro.
- a. Propositions I and III are correct.
 - b. Propositions II and III are correct.
 - c. Propositions II et IV are correct.
 - d. Propositions I et IV are correct.

Answer: b.

Question 23 (ch.10)

The law of one price is satisfied. Consider the following statements:

- I. The real exchange rate is equal to 1.
- II. There are arbitrage possibilities by selling in one country goods that were purchased in another country.
- III. The nominal exchange rate between two countries is equal to the ratio of price levels in the two countries.

Which of the following propositions is correct?

- a. Only propositions I and II are correct.
- b. Only propositions II and III are correct.
- c. Only propositions I and III are correct.
- d. All propositions are correct.

Answer: c

Question 24 (ch.11)

Which of the following causes a depreciation of the national currency?

- a. A decrease in taxes to foster economic growth.
- b. The introduction of a new duty on imports of goods and services.
- c. An increase in public transfers to the unemployed.
- d. A destabilization of the national economy caused by armed riots.

Answer: d.

Question 25 (ch.11)

An open economy imposes an import quota and reduces domestic taxes. What is the impact of these two policies on the real interest rate and the real exchange rate?

- a. The real interest rate increases and there is a depreciation of the domestic currency.
- b. The real interest rate decreases and there is a depreciation of the domestic currency.
- c. The real interest rate decreases and there is an appreciation of the domestic currency.
- d. The real interest rate increases and there is an appreciation of the domestic currency.

Answer: d)

Question 26 (ch.11)

Consider a world composed of two countries : A and B. If A faces economic woes following a political crisis, what are the consequences in the exchanges between A and B?

- a. Net capital outflows from B decrease and the real interest rate in A decreases.
- b. Net capital outflows from A decrease and the real interest rate in B increases.
- c. Net capital outflows from A increase and the real interest rate in B decreases.
- d. Net capital outflows from B increase and the real interest rate in A increases.

Answer: c)

Question 27 (ch.12)

An economy is at its long-run equilibrium. Which of the following events would necessarily lead to an increase in prices without modifying the long-run level of production?

- a. An increase in government expenditure.

- b. A decrease in investment.
- c. An increase in the wages negotiated by labor unions.
- d. The depletion of an oil field.

Answer: a.

Question 28 (ch.12)

Because of an earthquake, 25% of the production capacities of Seismicland are permanently destroyed. At the same time, the national government decides to increase its expenditures. What are the consequences for the national economy?

- a. The price level decreases, and the effect on long-term output is ambiguous.
- b. The long-term level of output decreases, and the price level increases.
- c. The long-term level of output does not change, and the price level increases.
- d. The price level decreases, and the long-term level of output increases.

Answer: b.

Question 29 (ch.13)

The Keynesian multiplier...

- a. ...is a positive function of the marginal propensity to import.
- b. ...is a negative function of the marginal propensity to consume.
- c. ...is a negative function of the income tax rate.
- d. ...is a positive function of the marginal rate of substitution.

Answer: c.

Question 30 (ch.13)

The crowding out effect...

- a. ... is due to the strong preference for liquidity that households have during recessions.
- b. ... is due to the increase in the interest rate following a shift to the right in aggregate demand that is caused by an expansive fiscal policy.
- c. ... is, in absolute value, necessarily smaller than the impact of the Keynesian multiplier on output when the government undertakes a restrictive fiscal policy.
- d. ... strengthens the impact of an expansive monetary policy on the level of output in the short run.

Answer: b)